

Chapter 7 - Part 1: Hypothesis testing for one sample

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3 Hypothesis testing for two independent samples

Outline

1 Review

2 Hypothesis testing for one sample

Generally, a hypothesis testing problem for the parameter θ will be formed in one of 3 following formulations (with θ_0 is a given value to be tested)

- $$\begin{cases} H_0 : & \theta = \theta_0, \\ H_1 : & \theta \neq \theta_0. \end{cases}$$

- $$\begin{cases} H_0: & \theta \geq \theta_0, (\text{ or } \theta = \theta_0) \\ H_1: & \theta < \theta_0. \end{cases}$$

- $$\begin{cases} H_0: & \theta \leq \theta_0, (\text{ or } \theta = \theta_0) \\ H_1: & \theta > \theta_0. \end{cases}$$

In the problem of hypothesis testing, we might make two mistakes as follows:

is the mistake in which one rejects H_0 although H_0 is actually true.

The probability of Type I error is α , the **significant level** of the test:

$$\alpha = \mathbb{P} \left(T \in W_\alpha \mid H_0 \text{ is true} \right) = \mathbb{P} \left(\text{reject } H_0 \mid H_0 \text{ is true} \right).$$

is the mistake in which one fails to reject H_0 although H_0 is false.

The probability of Type II error, denoted by β , is

$$\beta = \mathbb{P} \left(T \in W_{\alpha}^c | H_0 \text{ is false} \right) = \mathbb{P} \left(\text{not reject } H_0 | H_0 \text{ is false} \right).$$

Moreover, $(1 - \beta)$ is called **the power** of the test.

Type I error and type II error

Decision \ H_0	H_0 is true	H_0 is false
Fail to reject H_0	no error (1 - α)	Type II error β
reject H_0	Type I error α	no error (1 - β)

p -value = the smallest significant level that would lead to rejection of the null hypothesis H_0 .

Conclusion: reject H_0 if p -value $\leq \alpha$

The power of hypothesis testing

Definition 1.1

For a statistic of test $T = T(X_1, X_2, \dots, X_n; \theta)$ and a critical region W_α , the **power** of hypothesis testing is the probability of rejecting H_0 when H_0 is false, denoted by π ,

$$\begin{aligned}\pi &= 1 - \mathbb{P}(T \in W_\alpha^c | H_1) = \mathbb{P}(T \in W_\alpha | H_1) = 1 - \beta \\ &= \mathbb{P}(\text{reject } H_0 | H_0 \text{ is false}).\end{aligned}$$

A good statistical test comes with a strong power of hypothesis testing.

Outline

1 Review

2 Hypothesis testing for one sample

- Testing for the mean (expectation) μ of the population:
 - Case1: the variance σ^2 is known
 - Case2: The variance σ^2 is unknown and a large sample size
 - Case3: the variance σ^2 is unknown but with a small sample size
- Testing the proportion of the population

Contents:

- Testing for the mean (expectation) μ of the population:
 - In case the variance is known;
 - In case the variance is unknown and large sample size;
 - In case the variance is unknown but with small sample size.
- Testing for the proportion of the population.

1 Review

- Testing for the mean (expectation) μ of the population:

3 Hypothesis testing for two independent samples

Testing for the mean (expectation) μ of the population

Case1: the variance σ^2 is known

■ Assumptions:

- A random sample $X_1, X_2, \dots, X_n$ is chosen from the population following $\mathcal{N}(\mu, \sigma^2)$ with the unknown mean (expectation) μ .
 - The variance σ^2 is known.
 - Given a value μ_0 to be tested, it needs to compare μ with μ_0 .
- The considered hypothesis testing problem can be formed in one of the following 3 formulations, with a given significant level α ,

$$(a) \begin{cases} H_0 : \mu = \mu_0, \\ H_1 : \mu \neq \mu_0. \end{cases} \quad (b) \begin{cases} H_0 : \mu = \mu_0, \\ H_1 : \mu < \mu_0. \end{cases} \quad (c) \begin{cases} H_0 : \mu = \mu_0, \\ H_1 : \mu > \mu_0. \end{cases}$$

Case1: the variance σ^2 is known

Case1: the variance σ^2 is known

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Case1: the variance σ^2 is known

□ Calculate p -value corresponding to the alternative hypothesis H_1 and conclude as rejection of H_0 if

with a given significant level α .

- Formulas to calculate p -value are summarized in the Table 2.

Hypotheses	p - value
$H_0 : \mu = \mu_0$ $H_1 : \mu \neq \mu_0$	$p = 2 \cdot (1 - \Phi(z_0))$
$H_0 : \mu = \mu_0$ $H_1 : \mu < \mu_0$	$p = \Phi(z_0)$
$H_0 : \mu = \mu_0$ $H_1 : \mu > \mu_0$	$p = 1 - \Phi(z_0)$

Table 2: p -value corresponds to various formulations of H_1 .

Testing for the mean (expectation) μ of the population

Case1: the variance σ^2 is known - example 1

Example 2.1 (Two-sided test)

The manufacturing process of P/S toothpastes is designed to package toothpastes with the mean weight of 6 oz (1oz = 28gr). A sample of 30 toothpastes are randomly chosen for regular examination. The controller of this process must guarantee the weight of 6oz for each toothpaste; otherwise, if the weight is considerably either less or much than 6oz, the manufacturing process must be adjusted.

Suppose that the sample mean of 30 toothpastes is 6.1oz and the standard deviation of the population $\sigma = 0.2\text{oz}$.

With significant level 3%, is there evidence that the manufacturing process is working well?

Testing for the mean (expectation) μ of the populationCase1: the variance σ^2 is known - example 1 (cont.)Answer:

Let X be random variable presenting the weight of one toothpaste.

According to assumptions, $X \sim \mathcal{N}(\mu, 0.2^2)$. The procedure of hypothesis testing is as follows:

1. Formulate the hypotheses:

$$\begin{cases} H_0 : \mu = 6, \\ H_1 : \mu \neq 6. \end{cases}$$

2. Determine the level of significance: $\alpha = 0.03$.
3. From observed sample, calculate the statistic of test

$$z_0 = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{6.1 - 6.0}{0.2/\sqrt{30}} \approx 2.74;$$

4. Determine the critical region: reject H_0 if $|z_0| > z_{1-\frac{\alpha}{2}}$.

Testing for the mean (expectation) μ of the populationCase1: the variance σ^2 is known - example 1 (cont.)

Answer(cont.): We have $\alpha = 3\% = 0.03$ so $z_{1-\frac{\alpha}{2}} = z_{0.985} = 2.17$.

Thus, we will reject H_0 if

$$z_0 < -2.17 \quad \text{or} \quad z_0 > 2.17.$$

5. Conclusion: since $z_0 = 2.74 > 2.17 = z_{1-\frac{\alpha}{2}}$, we reject H_0 .

We conclude with a confident level $100 \times (1 - \alpha)\% = 97\%$ that the mean of weight of each toothpaste is not equal to 6oz.

■ Using p -value

4a. Calculate p -value in two-sided hypothesis testing problem:

$$p = 2 \cdot (1 - \Phi(|z_0|)) = 2 \cdot (1 - \Phi(2.74)) \approx 2 \cdot (1 - 0.9969) = 0.0062.$$

5a. Conclusion: with $\alpha = 0.03$, one has $p = 0.0062 < 0.03 = \alpha$ so we reject H_0 .
We conclude with a confident level $100 \times (1 - \alpha)\% = 97\%$ that the mean of weight of each toothpaste is not equal to 6oz.



Testing for the mean (expectation) μ of the population

Case1: the variance σ^2 is known - example 2

Example 2.2 (One-sided test)

Metro EMS: A hospital in the city center provides a home emergency medical service (EMS). With 20 ambulances, the objective is that the mean of time for a EMS is 12 minutes after urgent calls of request.

A random sample on response time of 40 emergency cases is selected and the sample mean is 13.25 minutes. Suppose that the standard deviation of the population is $\sigma = 3.2$ minutes.

The manager of EMS wants to carry out a hypothesis test, at a level of significance 5%, in order to determine that whether the response time of the home emergency medical service exceeds 12 minutes?

Testing for the mean (expectation) μ of the population

Case1: the variance σ^2 is known - example 2 (cont.)

Answer:

Let X be a random variable presenting the time response of the home emergency medical service.

According to assumptions, $X \sim \mathcal{N}(\mu, 3.2^2)$. The procedure of hypothesis testing is as follows

1. Formulate the hypotheses:

$$\begin{cases} H_0 : & \mu = 12, \text{ the response time is satisfactory;} \\ H_1 : & \mu > 12, \text{ the response time is NOT satisfactory.} \end{cases}$$

2. Determine the level of significance: $\alpha = 0.05$.
3. From observed sample, calculate the statistics of test

$$z_0 = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{13.25 - 12}{3.2/\sqrt{40}} \approx 2.47;$$

4. Determine the critical region: reject H_0 if $z_0 > z_{1-\alpha}$.

Testing for the mean (expectation) μ of the population

Case1: the variance σ^2 is known - example 2 (cont.)

Answer(cont.): We have $\alpha = 5\% = 0.05$ so $z_{1-\alpha} = z_{0.95} = 1.645$.

Thus, we shall reject H_0 if $z_0 > 1.645$.

5. Conclusion: since $z_0 = 2.47 > 1.645 = z_{1-\alpha}$ we reject H_0 .

We conclude with a confident level $100 \times (1 - \alpha)\% = 95\%$ that the Metro EMS does NOT satisfy the objective of that is the time for a EMS does not exceed 12 minutes.

■ Using p -value

4a. Calculate p -value in one-sided hypothesis testing problem:

$$p = 1 - \Phi(z_0) = 1 - \Phi(2.47) \approx 1 - 0.9932 = 0.0068.$$

5a. Conclusion: with $\alpha = 0.05$, one has $p = 0.0068 < 0.05 = \alpha$ so we reject H_0 .

We conclude with a confident level 95% that the Metro EMS does NOT satisfy the objective of that is the time for a EMS does not exceed 12 minutes.



Testing for the mean (expectation) μ of the population

Case2: the variance σ^2 is unknown and a large sample size

■ Assumptions:

- A random sample $X_1, X_2, \dots, X_n$ is chosen from the population following $\mathcal{N}(\mu, \sigma^2)$ with the unknown mean (expectation) μ and the variance σ^2 is unknown.
 $\Rightarrow$ use the sample variance S^2 as an estimate of σ^2 .
- The sample size is reasonably large: $n > 30$.
- Given a value μ_0 to be tested, it needs to compare μ with μ_0 .

■ The considered hypothesis testing problem can be formed in one of the following 3 formulations, at a given significant level α :

$$(a) \begin{cases} H_0 : \mu = \mu_0, \\ H_1 : \mu \neq \mu_0. \end{cases} \quad (b) \begin{cases} H_0 : \mu = \mu_0, \\ H_1 : \mu < \mu_0. \end{cases} \quad (c) \begin{cases} H_0 : \mu = \mu_0, \\ H_1 : \mu > \mu_0. \end{cases}$$

Testing for the mean (expectation) μ of the population

Case2: the variance σ^2 is unknown and a large sample size

▷ When the sample size is reasonably large, the random variable

$$Z_0 = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}, \quad (2)$$

can be approximately computed by the Gaussian distribution $\mathcal{N}(0, 1)$.

▷ Then, the critical region W_α or p -value are calculated similarly to the case 1 (in which the variance is known), except the replacement for $\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$ by $Z_0 = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$ in the formula (2).

Testing for the mean (expectation) μ of the population

Case3: the variance σ^2 is unknown but with a small sample size

■ Assumptions:

- A random sample $X_1, X_2, \dots, X_n$ is chosen from the population following the distribution $\mathcal{N}(\mu, \sigma^2)$ with the unknown mean (expectation) μ and the variance σ^2 is unknown.

$\Rightarrow$ use the sample variance S^2 to replace σ^2 .

- The sample size is small: $n \leq 30$.
- Given a value μ_0 to be tested, it needs to compare μ with μ_0 .

■ The considered hypothesis testing problem can be formed in one of the following 3 formulations,

at a given significant level α :

$$(a) \begin{cases} H_0 : \mu = \mu_0, \\ H_1 : \mu \neq \mu_0. \end{cases} \quad (b) \begin{cases} H_0 : \mu = \mu_0, \\ H_1 : \mu < \mu_0. \end{cases} \quad (c) \begin{cases} H_0 : \mu = \mu_0, \\ H_1 : \mu > \mu_0. \end{cases}$$

Testing for the mean (expectation) μ of the population
 Case3: the variance σ^2 is unknown but with a small sample size

Hypothesis testing procedure

1. Formulate the null hypothesis and the alternative hypothesis.
2. Determine the significant level α .
3. From the observed sample of size n , calculate the statistic for the test as

$$T_0 = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}, \quad (3)$$

where the random variable T_0 follows a t -Student distribution with $(n - 1)$ degree of freedom.

Testing for the mean (expectation) μ of the population
 Case3: the variance σ^2 is unknown but with a small sample size

Hypothesis testing procedure (cont.)

4. Determine the critical region W_α : summarized in the Table 3.

Hypotheses	The critical region
$H_0 : \mu = \mu_0$ $H_1 : \mu \neq \mu_0$	$W_\alpha = \left\{ t_0 : t_0 > t_{1-\frac{\alpha}{2}}^{n-1} \right\}$
$H_0 : \mu = \mu_0$ $H_1 : \mu < \mu_0$	$W_\alpha = \left\{ t_0 : t_0 < -t_{1-\alpha}^{n-1} \right\}$
$H_0 : \mu = \mu_0$ $H_1 : \mu > \mu_0$	$W_\alpha = \left\{ t_0 : t_0 > t_{1-\alpha}^{n-1} \right\}$

Table 3: The critical regions correspond to formulation of the alternative hypothesis H_1 (case of small sample size).

5. Conclusion: depending on the value of z_0 , we will
 reject H_0 or fail to reject H_0 .

Testing for the mean (expectation) μ of the population
 Case3: the variance σ^2 is unknown but with a small sample size

The use of p -value

□ Calculate p -value corresponding to the alternative hypothesis H_1 and conclude as rejection of H_0 if

$$p\text{-value} \leq \alpha,$$

with a given significant level α .

□ With T_{n-1} , a t -Student random variable with $(n - 1)$ degree of freedom, formulas for calculation of p -value are summarized in the [Table 4](#).

Hypothesis	p - value
$H_0 : \mu = \mu_0$ $H_1 : \mu \neq \mu_0$	$p = 2.\mathbb{P}(T_{n-1} \geq t_0)$
$H_0 : \mu = \mu_0$ $H_1 : \mu < \mu_0$	$p = \mathbb{P}(T_{n-1} \leq t_0)$
$H_0 : \mu = \mu_0$ $H_1 : \mu > \mu_0$	$p = \mathbb{P}(T_{n-1} \geq t_0)$

[Table 4](#): p -value corresponds to various formulations of H_1 (case of small sample size).

Testing for the mean (expectation) μ of the populationCase3: the variance σ^2 is unknown but with a small sample size - example 1

Example 2.3

On a highway, traffic police stations will periodically conduct speed examination at different locations. A random sample on speed of vehicles on this highway is chosen to test the following hypotheses

$$\begin{cases} H_0 : \mu = 65, \\ H_1 : \mu > 65. \end{cases}$$

The locations where data on vehicles' speed supports to reject H_0 are appropriate locations for placing the speed control radar.

At a location F , a sample on velocities of 64 vehicles that are examined randomly has the sample mean of 66.2 mph and the standard deviation 4.2 mph. Use $\alpha = 5\%$ to test these hypotheses.

Testing for the mean (expectation) μ of the population

Case3: the variance σ^2 is unknown but with a small sample size - example 1 (cont.)

Hypothesis testing procedure:

1. Formulate the hypotheses;
2. Determine the significant level $\alpha = 0.05$.
3. From observed sample of size n , since the variance σ^2 is unknown and the sample size $n = 64 \geq 30$ is reasonably large:

$$z_0 = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{66.2 - 65}{4.2/\sqrt{64}} = 2.286.$$

4. Determine the critical region: reject H_0 if

$$z_0 > z_{1-\alpha} = z_{0.95} = 1.645.$$

Testing for the mean (expectation) μ of the population

Case3: the variance σ^2 is unknown but with a small sample size - example 1 (cont.)

5. Conclusion: since $z_0 = 2.286 > 1.645 = z_{1-\alpha}$, we reject H_0 .

We conclude with confident level $100 \times (1 - \alpha)\% = 95\%$ the mean of vehicles' speed at the location F is greater than 65 mph. Therefore, the location F is appropriate for placing the speed control radars.

■ Using p -value

4a. Calculate p -value in one-sided-left hypothesis testing:

$$p = 1 - \Phi(z_0) = 1 - \Phi(2.286) \approx 0.0111.$$

5a. Conclusion: at a level of significance $\alpha = 0.05$, as $p = 0.0111 < 0.05 = \alpha$, we reject H_0 .

We conclude with confident level $100 \times (1 - \alpha)\% = 95\%$ the mean of vehicles' speed at the location F is greater than 65 mph. Therefore, the location F is appropriate for placing the speed control radars.

Testing for the mean (expectation) μ of the population

Case3: the variance σ^2 is unknown but with a small sample size - example 2

Example 2.4

A battery manufacturing company claims that the mean of lifespan of their produced batteries is at least 240 hours. Monitoring a random sample of 18 batteries to collect data as in the following

237	242	244	262	225	218
242	258	243	234	236	228
232	230	254	220	232	240

Suppose that the lifespan of batteries procedure by this manufacturer follows a normal distribution.

- Draw a stem-leafs graph for the data and comment.
- At a level of significance 5%, can one reject the claim on the manufacturer on their batteries' lifespan?

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- Testing for the mean (expectation) μ of the population:
 - Case1: the variance σ^2 is known
 - Case2: The variance σ^2 is unknown and a large sample size
 - Case3: the variance σ^2 is unknown but with a small sample size
- Testing the proportion of the population

3 Hypothesis testing for two independent samples

Testing the proportion of the population

■ Problem:

Consider a population X and let p be the unknown proportion of individuals possessing a property $\mathcal{A}$ in the population.

From a random sample $(X_1, X_2, \dots, X_n)$, at a given significant level α , test the hypotheses

$$(a) \begin{cases} H_0 : p = p_0 \\ H_1 : p \neq p_0 \end{cases} \quad \text{or} \quad (b) \begin{cases} H_0 : p = p_0 \\ H_1 : p < p_0 \end{cases} \quad \text{or} \quad (c) \begin{cases} H_0 : p = p_0 \\ H_1 : p > p_0 \end{cases}$$

where p_0 is a given value to be tested.

■ Assumptions:

When the sample size n is reasonably large, one can use the normal distribution to approximately compute for the binomial distribution, if

$$n \cdot p_0 \geq 5, \quad \text{and} \quad n \cdot (1 - p_0) \geq 5.$$

Testing the proportion of the population

- Observing an event $E =$ the occurrence of "individuals possessing the property $\mathcal{A}$ " in each of n times of independent experiments.

Let Y be the number of times that the event E occurs with n times of independent experiments, so $Y \sim B(n, p)$, and

$$\hat{P} = \frac{Y}{n},$$

is an unbiased estimator for p .

- Then, if H_0 is true, the statistic of test

$$Z_0 = \frac{\hat{P} - p_0}{\sqrt{\frac{p_0 \cdot (1 - p_0)}{n}}},$$

can be approximately computed by the Gaussian distribution $\mathcal{N}(0, 1)$ and we take Z_0 for testing of the hypotheses.

Hypothesis testing procedure

1. Formulate the null hypothesis and the alternative hypothesis.
2. Determine the significant level α .
3. Consider that H_0 is true, then calculate the statistic of test:

$$Z_0 = \frac{\hat{P} - p_0}{\sqrt{\frac{p_0 \cdot (1 - p_0)}{n}}}$$

Testing the proportion of the population

The use of p -value

Calculate p -value corresponding to the alternative hypothesis H_1 and conclude as rejection of H_0 if

$$p\text{-value} \leq \alpha,$$

with a given significant level α .

Formulas for calculation of p -value are summarized in the [Table 2](#).
(similar to the case 1 in hypothesis testing problem for the mean μ of the population)

Example 2.5

Use these data to test the announcement of the Traffic Safety Department at a significant level 5%.

1. Formulate the hypotheses: according to the announcement of the T.S.D., consider that $p_0 = 0.5$,

$$\begin{cases} H_0 & : p = p_0 = 0.5, \\ H_1 & : p \neq 0.5. \end{cases}$$

2. Determine the significant level $\alpha = 0.05$.

Answer:(cont.) Hypothesis testing procedure:

Testing the proportion of the population - example 2

Example 2.6

In the treatment of an infectious disease, a medicine is known to successfully treat 72% of infections. A new medicine is developed and trialed founding that it has successfully treated 42 out of 50 infected patients.

Is there efficiently strong evidence to conclude that the new medicine is more effective than the one medicine?

Calculate p -value.

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